

ECG Phase Estimator for Cardiac Artifact Removal

Extracted from: Cardiac Field Artifact Removal in Ear-EEG Using Repetitive Disturbance Rejection

Problem Formulation (Background)

Following repetitive-control modeling conventions, the measured ear-EEG signal is modeled as

$$y(t) = s(t) + d_r(t) + n(t), \quad (1)$$

where $s(t)$ is the neural signal of interest, $d_r(t)$ is the heartbeat-synchronous repetitive cardiac disturbance, and $n(t)$ aggregates asynchronous interference and sensor noise. The subscript r denotes the repetitive nature of the disturbance. Its period is heartbeat dependent and is set by ECG R-peak timing, with beat interval

$$T_k = t_{k+1} - t_k, \quad (2)$$

where t_k is the k -th detected R-peak instant.

Assumption 1. *The beat period T_k is assumed slowly varying, dominated by respiratory sinus arrhythmia (RSA), with additional small modulation from low-level muscle/motion effects in resting-state recordings.*

The repetitive disturbance is represented as a phase-indexed waveform

$$d_r(t) = d_r(\phi(t)), \quad \phi(t) = \frac{t - t_k}{T_k}, \quad t \in [t_k, t_{k+1}), \quad (3)$$

with $\phi(t) \in [0, 1)$ the R-peak-aligned cardiac phase.

Problem 1. *Estimate $\hat{t}_k$ robustly so that EEG samples can be phase-aligned via the phase model. The estimated cardiac phase between beats advances as*

$$\hat{\phi}(t) = \frac{t - \hat{t}_k}{\hat{T}_k}, \quad \hat{T}_k = \hat{t}_{k+1} - \hat{t}_k, \quad (4)$$

where $\hat{T}_k$ is the estimated beat period.

Note 1. *Since T_k is slowly varying (Assumption ??), $\hat{T}_k$ can be reliably predicted between successive R-peaks. The PI tracker advances phase continuously between beats so that $\hat{\phi}(t)$ remains well-defined in real-time causal operation.*

1 ECG Phase Estimator

1.1 Min/max tracker + PI/PID phase tracking

A QRS-emphasis band-pass feeds two streaming trackers: a peak tracker $pk(t)$ (fast attack, slow release) and a baseline tracker $bl(t)$ (slow moving average). The normalized gate

$$g(t) = \frac{x(t) - bl(t)}{pk(t) - bl(t)} \quad (5)$$

is amplitude-drift invariant, so posture/contact drift does not defeat detection. A Schmitt hysteresis and a 250 ms refractory yield clean beat gating; the fiducial t_k is the raw-reference maximum within the gated window (low jitter).

A PI phase-locked loop converts sparse fiducials into a continuous phase. Between beats the phase advances by the instantaneous rate estimate,

$$p(t+1) = [p(t) + \text{fr}(t)] \bmod 1, \quad \text{fr}(t) = \frac{1}{\text{RR}_{\text{est}}(t) f_s}, \quad (6)$$

where f_s is the sample rate. At each fiducial, a proportional term realigns phase and an integral term updates the rate estimate,

$$p \leftarrow p - K_p e_p, \quad (7)$$

$$\text{RR}_{\text{est}} \leftarrow \text{RR}_{\text{est}} + K_i (\text{RR}_{\text{meas}} - \text{RR}_{\text{est}}), \quad (8)$$

with e_p the wrapped phase error and K_p, K_i the PI gains. The loop bandwidth is set above respiratory sinus arrhythmia (RSA, ~ 0.2 – 0.4 Hz) and below per-beat detection noise. An arrhythmia supervisor classifies each beat by the ratio $\text{RR}_{\text{meas}}/\text{RR}_{\text{est}}$ and applies the matching reflex—*premature*: trust the measured phase, freeze the rate and learning; *missed*: trust the prediction, keep the rate—protecting both the loop and the template.

1.2 Algorithm summary

Per sample: (1) update trackers (??) and gate; (2) advance phase (??), correct at fiducials (??)–(??), run the supervisor; (3) ILC lookup, limited subtraction, and template update on normal beats. The proposed online path is therefore constant-time per sample, with template memory linear in the window length.

2 Complexity Analysis

Let L_w be the template length and C the channel count. The proposed phase-tracker + ILC path is lightweight: tracker updates and gating are constant-time, PI/PID phase update is constant-time (amortized), and ILC lookup/update is constant-time with memory L_w . Thus, per-sample complexity is $O(1)$ and memory is $O(L_w)$ for each channel. This profile is suitable for always-on low-power ear-EEG devices. For fairness, baseline adaptive cancellers (e.g., LMS/RLS) are reported separately in experiments and are not part of the proposed online deployment path.

[width=]fig_{real}/fig_{real}_coherence_{combined}.png

Figure 1: Real ear-EEG/ECG coherence. (a) Whole-record Welch estimate (0–60 Hz): indistinguishable from zero across the spectrum. (b) Beat-synchronous estimate (8–20 Hz band), pooled over rolling 12-beat batches across a 10-minute recording (715 estimates): coherence stays low and stable even under best-case R-locked alignment.

3 Experiments

3.1 Real ear-EEG validation: coherence and repeatable-template evidence

To test the coherence assumption behind reference-driven cancellers and confirm that a genuine repeatable cardiac template exists outside simulation, we analyze a 10-minute segment (717 beats; 2 premature, 0 missed) from the SeizeIT2 dataset [?], which pairs a wearable behind-the-ear EEG channel (256 Hz) with a synchronized chest ECG lead during video-EEG epilepsy monitoring.

Coherence is near zero, even under best-case R-locked pooling. Fig. ??(a) shows magnitude-squared coherence between the ECG reference and the ear-EEG channel from long, many-average Welch segments across 0–60 Hz: coherence averages 6×10^{-10} over 8–20 Hz and is visually indistinguishable from zero across the whole spectrum – inconsistent with any stationary linear channel between $r(t)$ and $d(t)$. A skeptic could object that this washes out a relationship holding only locally around each beat; pooling short-time estimates over rolling 12-beat R-aligned batches (715 estimates, Fig. ??(b)) gives the same answer – coherence stays low and stable (0.06–0.17, mean 0.11) throughout the recording, an order of magnitude below what a usable Hammerstein/RLS reference-driven canceller would need. Even granting the reference every possible timing advantage, the ear-EEG artifact waveform is not linearly predictable from the ECG waveform.

A repeatable, R-anchored template nonetheless exists. Coherence measures waveform predictability from the reference, not repeatability across beats. Non-causal ILC/synchronous averaging (§IV-B) applied directly to the real ear-EEG channel (Fig. ??) recovers a clear, consistent deflection time-locked to the ECG R-peak, confirming the cardiac field artifact at the ear is genuinely timing-repeatable – exactly the property ILC exploits – even though it fails the coherence test above. This is the real-data analogue of synchronous-average disk-servo RRO/NRRO separation: only the R-peak *instant* is used, never the ECG waveform shape. The causal-vs-buffered R-spike tradeoff is likewise real, not a simulation artifact: buffered matches non-causal (R-spike peak-to-peak 5.02×10^{-5} V vs. 5.07×10^{-5} V), while the causal template is capped by RR-prediction error to 3.47×10^{-5} V, a 32% reduction, consistent with §IV-B. Cardiac-band (1–20 Hz) power is reduced by 0.9/0.8/0.4 dB for non-causal/buffered/causal respectively; the modest magnitude versus the semi-synthetic ceiling (Table ??) reflects broadband physiological noise at this electrode and that only the beat-locked component is addressable by ILC by construction.

These observations are the empirical basis for this paper’s scope: a null coherence result that rules out a reference-driven residual stage as a primary contribution here, and a positive repeatability result that justifies keeping the timing-driven ILC stage. The linear RLS canceller in Table ?? is retained only as a semi-synthetic baseline, evaluated on data explicitly given a known Hammerstein/volume-conduction channel between $r(t)$ and $d(t)$; it is not part of the proposed real-time deployment path.

3.2 Semi-synthetic data

The real-data evidence above has no ground truth $s(t)$, so we complement it with a semi-synthetic protocol that provides exact-reference SNR metrics; the ILC-driven design validated qualitatively

[width=0.85]fig_{real}/fig_{real_causal_buffered}.png

Figure 2: Non-causal, buffered, and strict-causal templates learned from real ear-EEG (with one ECG beat shown only for R-peak timing reference). A clear, repeatable R-locked deflection is recovered despite the near-zero coherence of Fig. ?? . Buffered matches non-causal; the causal (zero-latency) template’s R-spike is measurably damped by RR-prediction error, confirming the latency/suppression tradeoff of §IV-B on physiological data.

[width=0.85]fig1_{convergence_spectrum}.pdf

Figure 3: (a) Online SNR convergence for the three modes (single causal pass). (b) Power spectrum before/after cleaning (buffered); shaded band marks the preserved 10 Hz alpha rhythm.

above is here validated quantitatively. Following the semi-synthetic protocol common in EEG denoising [?], a low-CFA neural surrogate (pink noise plus an amplitude-modulated 10 Hz alpha rhythm) provides ground truth $s(t)$. A synthetic ECG source $u(t)$ with RSA-modulated RR intervals is converted into a beat-locked artifact template with slow amplitude and morphology drift, then added at a controlled input SNR; one premature and one dropped beat are injected. This yields exact-reference metrics without manual labeling. Metrics: artifact SNR (steady state), spectral cardiac-peak reduction, and alpha preservation.

3.3 Baselines

Within the semi-synthetic protocol only (§?? shows why a reference-driven residual stage cannot contribute on the real recording), we compare against (i) a linear RLS/LMS ECG-reference canceller [?, ?, ?]; (ii) an EKF/state-space canceller [placeholder—to be added] [?]; and (iii) low-channel ICA [placeholder—to be added] [?], which is expected to degrade at $C < 8$.

3.4 Results

Table ?? reports steady-state artifact SNR for the main approaches, generated directly by the companion script to keep manuscript numbers synchronized with code. Fig. ?? shows single-pass online convergence for the three operating modes and the corresponding buffered-mode spectrum before and after cleaning. In single-pass online operation, buffered mode converges faster and to a higher final SNR than causal mode, consistent with the expected one-beat anchoring advantage. Spectrally, the cardiac comb is strongly reduced after cleaning while the 10 Hz alpha peak is preserved within about 10%, an implicit heartbeat-evoked-potential safeguard [?]. Ablations show that phase anchoring quality and mode selection (non-causal vs buffered vs strict-causal) dominate performance: buffered ILC is closest to non-causal performance, while strict-causal ILC is limited by RR prediction accuracy around sharp R-wave transitions.

Fig. ?? shows the time-domain cancellation over three consecutive beats in the converged regime. The synthetic reference $r(t)$ exhibits full PQRST morphology—P wave, QRS complex, and T wave—rather than a simplified sinusoidal surrogate, so the canceller must contend with a sharp, high-amplitude R deflection and a low-amplitude pre-R P wave. After processing, the cleaned EEG closely tracks the ground-truth neural signal $s(t)$: the R spike is suppressed by roughly an order of magnitude, the QRS complex is removed, and the residual P wave is attenuated, while the ongoing alpha oscillation is retained. This confirms that the phase-anchored template and residual canceller act on realistic cardiac morphology, not a reduced model.

[width=0.85]fig2 t imedomain.pdf

Figure 4: Time-domain cancellation over three beats (converged, buffered mode). (a) Synthetic ECG reference $r(t)$ with full PQRST morphology; vertical lines mark detected R fiducials t_k . (b) Contaminated EEG $y(t)$, cleaned output, and ground-truth neural signal $s(t)$ over the same window.